



BUS-401: TQM Tools and Techniques II

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Concept of Quality Circle

Quality Circle is a small group of employees in the same work-area or doing a similar type of work who voluntarily meet regularly for about an hour every week to identify, analyze and resolve work-related problems, leading to improvement in their total performance, and enrichment of their work life” (Udupa 1986)

Quality circles are a formal, institutionalized mechanism for productive and participative problem-solving interaction among employees” (Lozano & Thompson 1980)

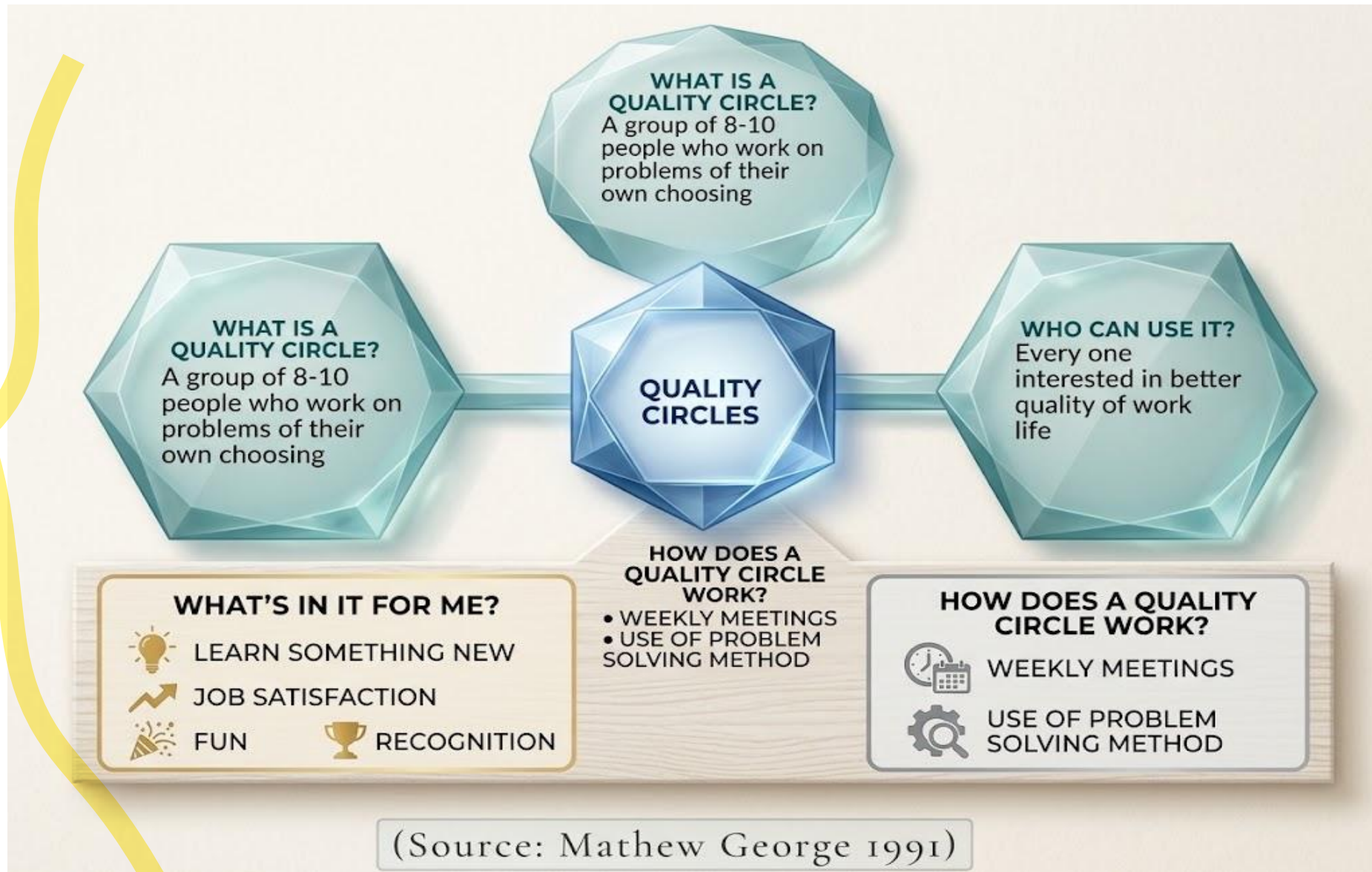
Quality control circle is not just a little room adjacent to the factory floor, whose occupants make a nuisance of themselves to everyone else. It is a state of mind and a matter of leadership with everyone from the president to production trainee involved” (Rehder 1981)

Features of Quality Circle



- ✓ Quality circle is a small group of employees
- ✓ Quality circle is organized in the same work area or doing similar type of work
- ✓ Quality circles are voluntary
- ✓ Quality circles meet regularly for about an hour every week
- ✓ Quality circles identify, analyses and resolves work-related problems
- ✓ Quality circle leads to total performance
- ✓ Quality circle enriches work life

Quality Circle in a nutshell is presented as-



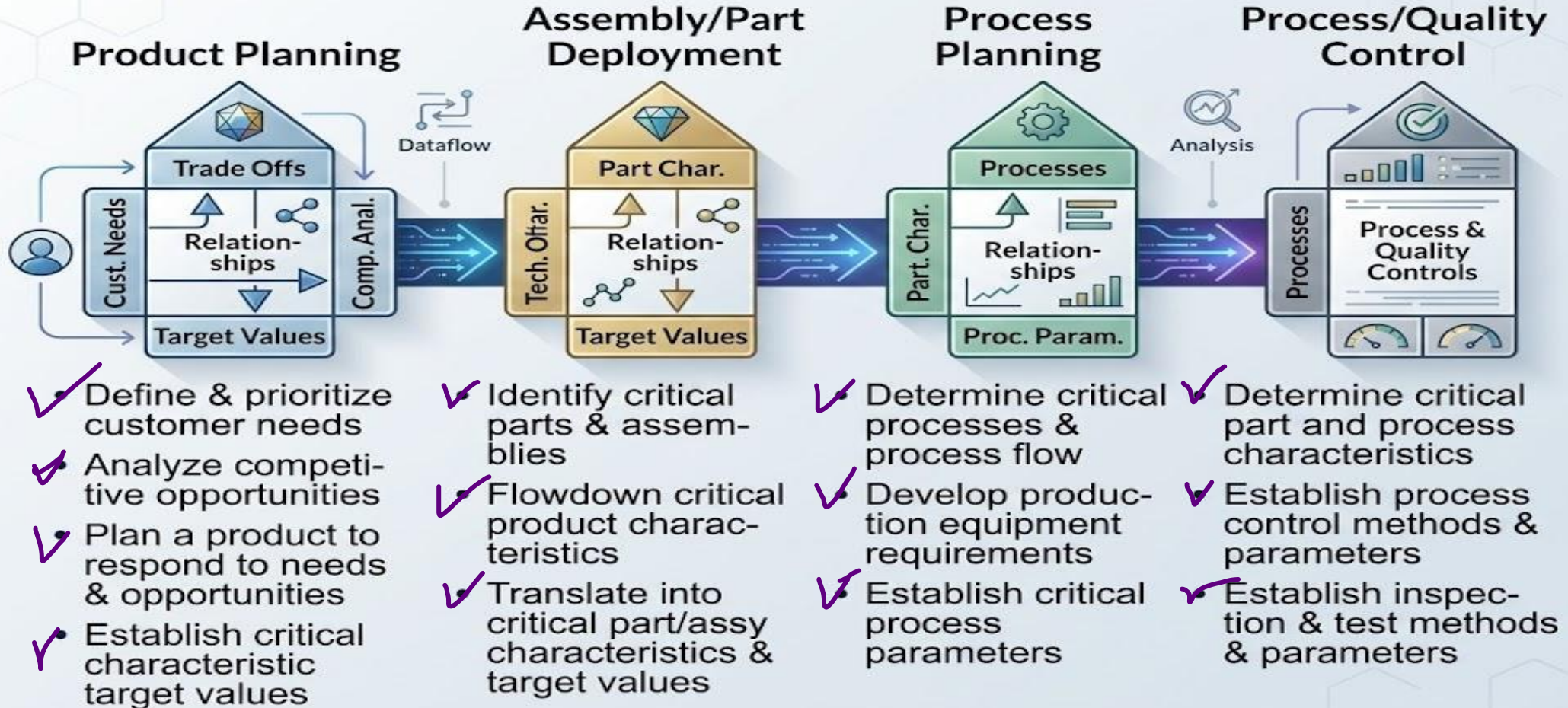
Quality Function Development



- ☐ Quality Function Deployment (QFD) is a structured approach to defining customer needs or requirements and translating them into specific plans to produce products to meet those needs.
- ✓ ☒ The "voice of the customer" is the term to describe these stated and unstated customer needs or requirements.
- ✓ ☒ The voice of the customer is captured in a variety of ways: direct discussion or interviews, surveys, focus groups, customer specifications, observation, warranty data, field reports, etc.
- ☐ QFD is oriented toward involving a team of people representing the various functional departments that have involvement in product development: Marketing, Design Engineering, Quality Assurance, Manufacturing/ Manufacturing Engineering, Test Engineering, Finance, Product Support, etc.

QFD METHODOLOGY FLOW

FOUR-PHASE QFD APPROACH



Project Management(PM) concept

- ✓ *"A project is a one-shot, time-limited, goal-directed, major undertaking, requiring the commitment of varied skills and resources".*
- ✓ Project management is concerned with the overall planning and co-ordination of a project from conception to completion aimed at meeting the stated requirements and ensuring completion on time, within cost and to required quality standards.

Project management is normally reserved for focused, non-repetitive, time-limited activities with some degree of risk and that are beyond the usual scope of operational activities for which the organization is responsible.

Steps in Project Management

- ✓ i. **Project Definition and Scope**
- ✓ ii. **Technical Design**
- ✓ iii. **Financing**
- ✓ iv. **Contracting**
- ✓ v. **Implementation**
- ✓ vi. **Performance Monitoring**

PM concept



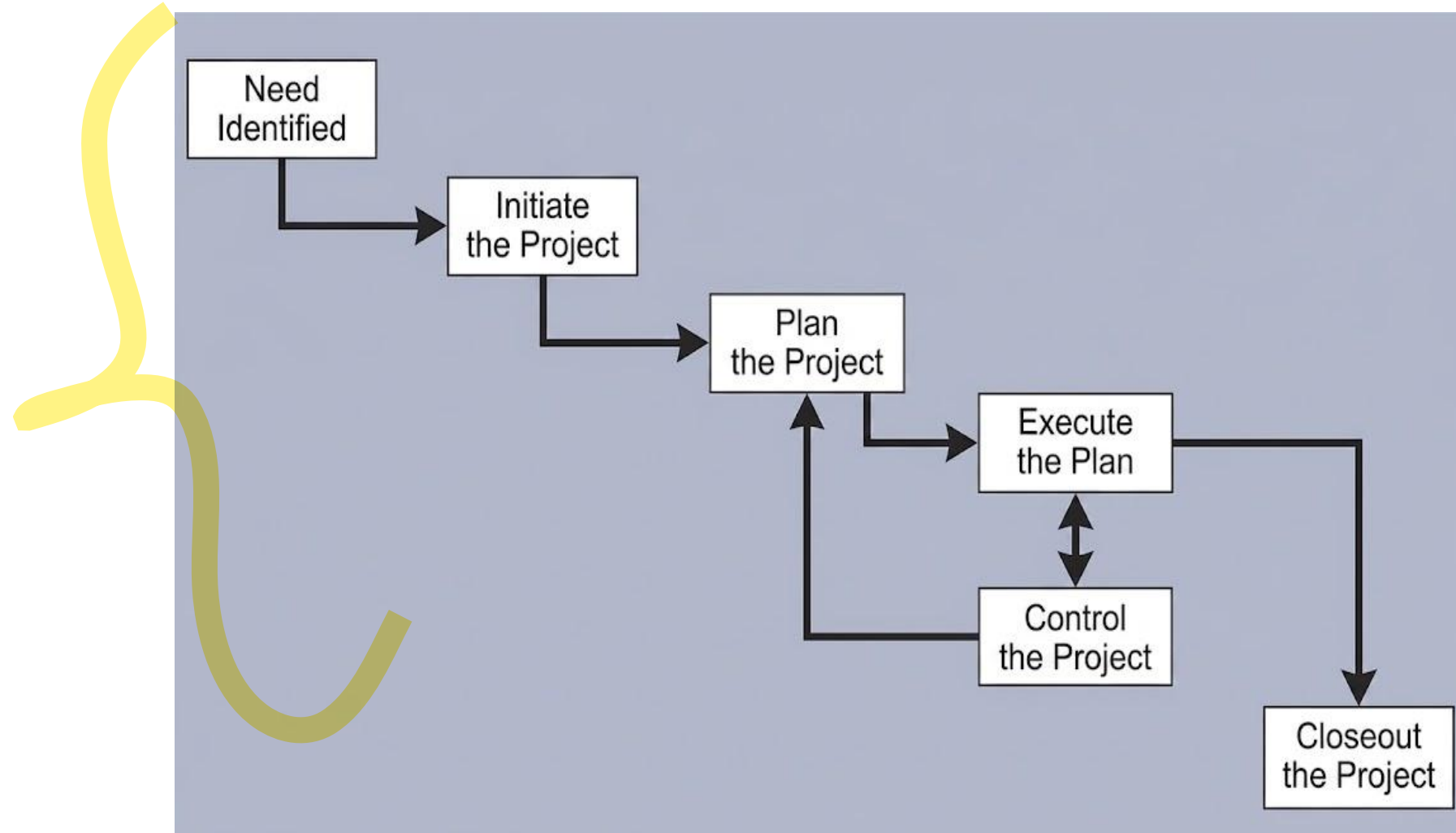
Four Basic Elements of Project Management

- i. Resource
- ii. Time
- iii. Cost
- iv. Setup

How the 80/20 Rule can help a project manager?

The 80/20 Rule means that in anything a few (20 percent) are vital and many (80 percent) are trivial. Successful Project Managers know that 20 percent of the work (the first 10 percent and the last 10 percent) consumes 80 percent of your time and resources.

Project Management Life Cycle

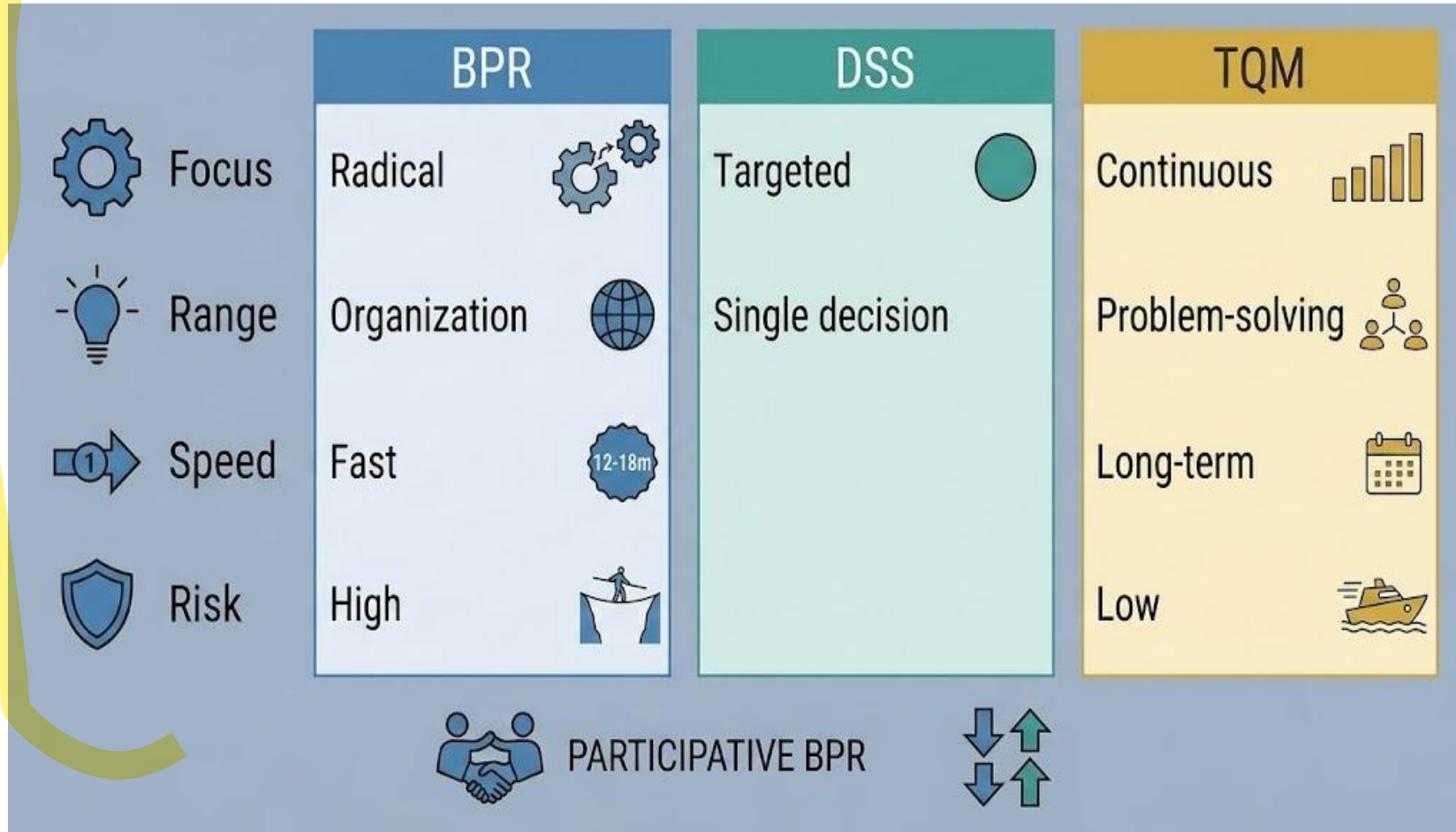


BPR- BUSINESS PROCESS REENGINEERING



- ❑ The aim of BPR, according to Sherwood-Smith (1994), is “seeking to devise new ways of organizing tasks, organizing people and redesigning IT systems so that the processes support the organization to realize its goals”.
- ❑ *[Reengineering is] the fundamental rethinking and radical redesign of business processes to achieve dramatic improvements in critical, contemporary measures of performance, such as cost, quality, service and speed.*
- ❑ Hammer and Champy (1993) define a process as: *a collection of activities that takes one or more kinds of input and creates an output that is of value to the customer.*
- ❑ *A process is structured change, i.e. there is a pattern of events which an observer may recognize across different actual examples (or occurrences) of the process, or which may be made manifest, or implemented, in many different occurrences. In BPR, the process to be reengineered is the so-called business process.*

BPR, DSS and TQM



BPR drives radical, organization-wide changes for fast high-risk gains, whereas DSS focuses specifically on individual decision-making. In contrast, TQM relies on low-risk, gradual, bottom-up improvements, which can be combined with BPR to create a less risky "participative BPR" approach.

Taguchi's Approach to Quality

- ❑ Taguchi defines quality in a negative way as 'loss imparted to society from the time the product is shipped'.
- ❑ Professor, Genichi Taguchi, director of the Japanese Academy of Quality and four times recipient of the Deming Prize, devised a quality improvement technique that uses experimental design methods for efficient characterization of a product or process, combined with a statistical analysis of its variability.
- ❑ Taguchi's main objective lies in his definition of quality (or non-quality):
'Quality is the amount of damage incurred to society from the moment the product has left the factory.'

Conventional Loss function and Taguchi's Loss Function

The traditional quality metric suggests

- ✓ All products within specifications are equally good
- ✓ All products outside specifications equally bad.

Unfortunately, this definition has led to the mindset which becomes a barrier towards improvement in products.

Figure 1 shows the target value, lower specification limit (LSL) and upper specification limit (USL). Both the products A and B are considered equally good although product A meets the specifications as per the target and product B is within the specification limit but far away from the target. Product C is considered to be non-acceptable or equally unacceptable as it is beyond the lower specification limit.

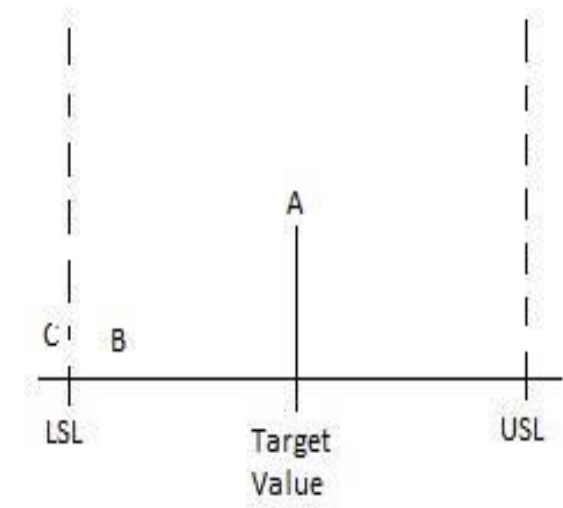


Figure 1: Traditional Loss Metric

Conventional Loss function and Taguchi's Loss Function

It was Taguchi who defined and quantified quality loss via his '**Loss Function**' which is quite different from the traditional concept. He unites the financial loss with the functional specification through a quadratic relationship that comes from a **Taylor Series Expansion** (which is an approximation method).

Taguchi's loss function is as under:

$$L(Y) = \frac{M}{D^2} (Y-t)^2$$

Where $L(Y)$ = the quality loss function

t = target value/most desirable value of a product

Y = performance of a product/the response found in test or analysis

M = producer's loss (in monetary terms)

D = customer's tolerance = $USL - t$

Conventional Loss function and Taguchi's Loss Function

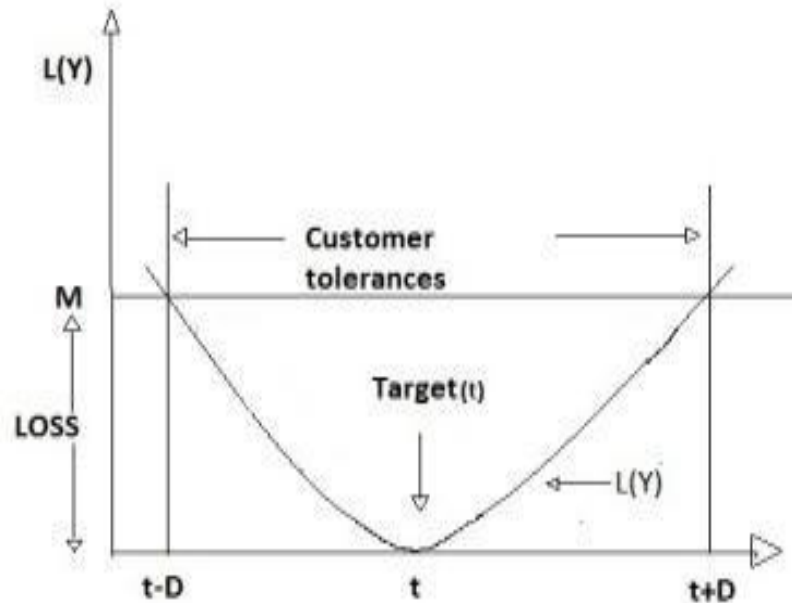


Figure 2: Graphical representation of Taguchi's Loss Function

The figure 2 shows the Taguchi's loss function

As per figure 2 the loss to the society increases as the performance characteristic deviates from the target value (t). In general, this loss is square of the deviation from the target. In the figure $t + D$ and $t - D$ are the functional limits beyond which 50% of system product needs customer maintenance.

A leather manufacturer produces upper leather with a specified thickness. According to the company's conventional specification system :

Lower Specification Limit (LSL) = 1.10 mm

Target thickness = 1.20 mm

Upper Specification Limit (USL) = 1.30 mm

The company considers any leather between 1.10 and 1.30 mm to be acceptable. However, the production manager has noticed that leather pieces having thicknesses close to 1.10 or 1.30 mm often create problems during cutting, stitching and lasting, even though they technically pass the specification. The company estimates that when leather reaches either specification limit, the resulting quality-related cost to the company/customer is Tk 200 per piece. During inspection, five leather pieces are found with the following thicknesses :

Piece A = 1.20 mm

Piece B = 1.25 mm

Piece C = 1.28 mm

Piece D = 1.30 mm

The quality manager wants to determine whether all four pieces should really be considered equally good.

Questions:

- ✓ a. Identify the target value, tolerance, and maximum loss**
- ✓ b. Calculate the quality loss for each leather piece.**
- ✓ c. Which piece has the highest quality loss?**
- d. Explain why the conventional specification approach and Taguchi's approach give different conclusions.**

Solution

✓ (a) $t = 1.20 \text{ mm}$

$$D = 1.30 - 1.20 = 0.10 \text{ mm}$$

$$M = 200 \text{ Tk}$$

✓ (b) Therefore, $L(Y) = \frac{M}{D^2} (Y-t)^2$

$$\text{So, } L(Y) = \frac{200}{0.10^2} (Y-1.20)^2 = 20,000 (Y-1.20)^2$$

✓ (c) For Piece A = 1.20 mm,

$$L(1.20) = 20,000 (1.20-1.20)^2 = 0$$

Similarly,

$$\text{For, Piece B} = 1.25 \text{ mm, } L(1.25) = 5 \text{ tk}$$

$$\text{For, Piece C} = 1.28 \text{ mm, } L(1.28) = 128 \text{ tk}$$

$$\text{For, Piece D} = 1.30 \text{ mm, } L(1.25) = 200 \text{ tk}$$

Leather piece	Thickness(mm)	Decision	Taguchi Loss (Tk)
A	1.20	Accept	0
B	1.25	Accept	5
C	1.28	Accept	128
D	1.30	Accept	200

Under the conventional quality system, all four pieces are acceptable because they fall within 1.10–1.30 mm.

Taguchi's Loss Function shows that they are not equally good.

Piece A is closest to the target and produces no loss, while Piece C, although still technically acceptable, has a substantial estimated loss of **128 tk**.

This demonstrates Taguchi's central principle:

Quality should not be viewed simply as “within specification” or “outside specification”; quality loss increases as the product moves away from the target.